SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B. E / B.Tech** Name: **A. Midhuna**

Course Code: **ITA0513** Reg No: **192521302**

Course Name: **Computer Vision**

Duration: **1:30 Hr**

Date: **2/11/26** Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.
CO2	Apply image enhancement and segmentation techniques for extracting meaningful regions from images.

Question-wise Mark Distribution

Question	Marks	CO	Bloom's Level
1 (a)–(d)	10	CO1	BL3
2 (a)–(d)	10	CO1	BL3
3 (a)–(b)	5	CO1	BL3
3 (c)–(d)	5	CO2	BL3
4 (a)–(d)	10	CO2	BL3
5 (a)–(d)	10	CO2	BL3
Total	50	–	–

CO-wise marks: CO1 = 25 marks (50%); CO2 = 25 marks (50%); Total = 50 marks.

Answer All Questions

1. A highway toll plaza is installing an Automatic Number Plate Recognition (ANPR) system. ANPR is a camera-based system that photographs the number plate of each passing vehicle and reads the registration number automatically using Optical Character Recognition (OCR) software, so that the toll is charged without stopping the vehicle. The camera uses an area sensor (a two-dimensional grid of light-sensing elements) of 1920 × 1080 pixels, and its horizontal field of view covers the full 3.84 m width of the lane. Vehicles pass at up to 72 km/h. Each number plate is 500 mm wide, with characters 65 mm high and 10 mm stroke (line) thickness. The OCR software reads correctly only if a character is at least 20 pixels high and its stroke is at least 2 pixels thick. The illumination is i = 1000 units by day and i = 200 units at night (from an infrared lamp). The reflectance (fraction of incident light reflected) is 0.85 for the white plate, 0.05 for the black characters and 0.30 for a mud stain.

(a) Using the image formation model $f(x, y) = i(x, y) \cdot r(x, y)$, compute the day and night intensities of the plate, characters and mud stain, and the character-to-plate contrast (difference). (3)

(b) Determine the sampling interval in mm/pixel and the plate width, character height and stroke thickness in pixels, and validate the OCR requirements. (3)

(c) Motion blur is the smearing caused by vehicle movement during the exposure time. Compute the blur in pixels for a 1 ms exposure and the longest exposure that keeps the blur within one pixel. (2)

(d) Justify an area sensor with a global shutter (all pixels exposed at the same instant) over a single sensor or a line-scan sensor (one row of elements), and interpret the effect of night-time contrast on character reading. (2)

(10) (CO1) (BL3)

2. A government office installs a self-service kiosk that scans citizens' identity cards and uploads them to an online archive. Each card measures 3.0 inch × 2.0 inch and is scanned in grayscale. The sampling density is expressed in dots per inch (dpi), that is, the number of pixels captured along one inch of the card. The thinnest printed line on the card is 0.25 mm wide, and the text-reading software needs every such line to be covered by at least 2 pixels. The kiosk scans 120 cards per minute and sends the uncompressed images over a network link of 10 Mbps (megabits per second). Sampling densities of 150, 300 and 600 dpi and bit depths of 8, 4 and 2 bits per pixel are being compared. Five sample pixels from the photograph on the card have intensities 12, 75, 130, 200 and 250. Take 1 inch = 25.4 mm, 1 KB = 1024 bytes and 1 Mbps = 10^6 bits per second.

(a) Compute the image size in pixels and the 8-bit storage in bytes and KB at 300 dpi and at 600 dpi. (3)

(b) Determine the number of pixels across the thinnest line at 150 dpi and 300 dpi, and the 8-bit data rate needed at 300 dpi and 600 dpi. Validate each setting. (3)

(c) Apply uniform 2-bit quantization (4 levels with step 64 and output levels 0, 85, 170 and 255) to the five samples and compute the mean squared quantization error. (2)

(d) Recommend the sampling density and bit depth for the kiosk, and justify your choice using (a)–(c) and the effect of false contouring (visible false bands caused by too few gray levels). (2)

(10) (CO1) (BL3)

3. A ceramic tile factory uses a camera to find hairline cracks (very thin surface cracks) in glazed tiles. The camera gives 8-bit grayscale images in which one pixel covers 0.5 mm of the tile. Pixel positions are written as (row, column). In one image, a straight, one-pixel-wide diagonal crack runs from p = (2, 3) to q = (6, 6), and the crack pixels touch each other mostly at their corners. Crack pixels have intensities between 20 and 40, while the good glaze has intensities between 180 and 220. Before thresholding, the engineer must measure the crack length and then increase the crack-to-glaze contrast using either (i) a power-law (gamma) transformation $s = 255 \times (r / 255)^{\gamma}$ with $\gamma = 0.5$, or (ii) a linear contrast stretch that maps the input range [20, 220] to [0, 255].

(a) Compute the Euclidean, city-block (D4) and chessboard (D8) distances between p and q in pixels and in millimetres, and identify the measure that gives the true crack length. (3)

(b) List the 4-neighbours N4(p) and the diagonal neighbours ND(p), and justify whether 4-, 8- or m-adjacency should be used to trace the crack as one connected path. (2)

(c) Apply both transformations to the intensities 20, 40, 180 and 220. (3)

(d) Compute the contrast (lowest glaze value − highest crack value) before and after each transformation, and justify the better option. (2)

(a)–(b): (5) (CO1) (BL3) (c)–(d): (5) (CO2) (BL3)

4. An oil company inspects the welded joints of an undersea pipeline using a Remotely Operated Vehicle (ROV), an unmanned underwater robot that carries a camera and a lamp and is controlled from a ship. Sea water absorbs and scatters light, so the images are dark and low in contrast, and small corrosion pits near the weld are hard to see. A 3-bit image region (L = 8 gray levels, 0 to 7) of 8 × 8 = 64 pixels has the histogram below. The enhancement must be automatic, because the operator cannot tune parameters for every frame.

Intensity rk	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Pixel count nk	4	10	16	14	8	6	4	2

(a) Compute the normalised histogram pr(rk) = nk / 64 and the cumulative distribution function (CDF) for each level. (3)

(b) Apply histogram equalization to find the new level sk of each input level, and construct the equalized histogram. (3)

(c) Compute the mean and standard deviation of the intensities before and after equalization, and interpret the change in contrast. (2)

(d) Compare global histogram equalization, linear contrast stretching and adaptive histogram equalization (equalization done separately in small tiles of the image), and justify the best method when the ROV lamp lights the scene unevenly. (2)

(10) (CO2) (BL3)

5. A Printed Circuit Board (PCB) factory uses Automated Optical Inspection (AOI), a camera system that checks every board automatically, to measure solder pads (the shiny metal areas where component leads are soldered). Under a ring light, the board appears dark, the solder pads appear bright, and flux residue (a chemical left over from soldering) gives medium gray values. The 3 × 3 patch I below is taken at the boundary of a pad; a pixel is marked as an edge if its gradient magnitude exceeds T = 200. Matrices are written row by row, with rows separated by semicolons. Sobel masks: Gx = [−1 0 1; −2 0 2; −1 0 1], Gy = [−1 −2 −1; 0 0 0; 1 2 1]. Prewitt masks: Gx = [−1 0 1; −1 0 1; −1 0 1], Gy = [−1 −1 −1; 0 0 0; 1 1 1].

I =

12	15	85
10	14	90
8	12	105

Nine pixels sampled across the board have intensities 22, 28, 30, 36, 64 (flux residue), 170, 185, 195 and 210.

(a) Apply the Sobel masks to compute Gx, Gy, the gradient magnitude and the gradient direction at the centre pixel; decide whether it is an edge and state the edge orientation. (3)

(b) Apply the Prewitt masks, compare the magnitude with the Sobel result, and justify the better operator for noisy images. (2)

(c) Apply the iterative global thresholding algorithm (start with T = mean intensity; stop when the change in T is less than 0.5), showing each iteration, and classify the pixels as pad or board. (3)

(d) Interpret how the flux residue is classified, and justify when Otsu's method or region growing should replace basic global thresholding. (2)

(10) (CO2) (BL3)

CIA - 1

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Course : ITA0513 Computer vision

1. ANPR System - Image processing problem

a) Gin:

Image formation model:

$$f(x,y) = i(x,y) \times r(x,y)$$

where

i = illumination

r = Reflectance

illumination: Day = 1000 units

Night = 200 units

Reflectance: white plate = 0.85

Black characters = 0.05

mud stain = 0.30

	Day time intensity:	Night time intensity
white plate	$f = 1000 \times 0.85 = 850$	$f = 200 \times 0.85 = 170$
Black characters	$f = 1000 \times 0.05 = 50$	$f = 200 \times 0.05 = 10$
mud stain:	$f = 1000 \times 0.30 = 300$	$f = 200 \times 0.30 = 60$

Day: $850 - 50 = 800$ Night: $170 - 10 = 160$

b) Gin: Sensor resolution = 1920 x 1080 pixels

Horizontal field = 3.84m = 3840mm

Plate width = 500mm

character height = 65mm

thickness = 10mm

Sampling interval: $= \frac{\text{Field of view}}{\text{Horizontal pixels}} = \frac{3840}{1920} = 2\text{mm/pixel}$

plate width pixels $= \frac{500}{2} = 250$

character height $= \frac{10}{2} = 5$

Stroke thickness $= \frac{65}{2} = 32.5$

c) Gin:

vehicle speed = 72km/h

Exposure time = 1ms = 0.001S

Sampling interval = 2mm/pixel

Convert speed $= 72 \times \frac{1000}{3600} = 20\text{m/s}$

Calculate motion blur 1ms: $20 \times 0.001 = 0.02\text{m} = 20\text{mm}$

$= 20/2 = 10$ pixels

calculate maximum one pixel blur 2mm = 0.002m

$= \frac{0.002}{20} = 0.0001\text{ sec} = 0.1\text{ms}$

d) An area sensor with a global shutter captures the complete number plate image at the same instant avoiding motion distortion. A single sensor cannot capture a complete image while line scan sensor captures one row time may cause distortion moving vehicle.

Day time contrast = 800 units night time contrast = 160 units

Lower night time contrast makes characters harder recognize. Therefore infrared illumination needed to improve OCR accuracy

Soln

a) calculate image size

Gn: card size : 3inch x 2inch

Bit depth : 8 bits

Sampling density : 300 dpi and 600 dpi

At 300 dpi

width : $3 \times 300 = 900$ pixels

Height : $2 \times 300 = 600$ pixels

Total pixels $900 \times 600 = 540000$

Storage $\frac{540000 \times 8}{8} = 540000$ bytes

$\frac{540000}{1024} = 527.34$ KB

At 600 dpi

$= 3 \times 600 = 1800$ pixels

$= 2 \times 600 = 1200$ pixels

$= 1800 \times 1200 = 2160000$

$= \frac{2160000 \times 8}{8} = 2160000$ bytes

$\frac{2160000}{1024} = 2109.375$ KB

b) calculate pixels across thinst

Thinnest = 0.25mm 1inch = 25.4mm,

Minimum = 2 pixels

At 150 dpi

pixels $= \frac{0.25}{25.4} \times 150$

$= 1.476$ pixels

At 300 dpi

pixels $= \frac{0.25}{25.4} \times 300$

$= 2.953$ pixels

cards per min = 120

Data rate $= \frac{540000 \times 8 \times 120}{60} = 8640000$ bits/s $= 8.64$ mbps

Data rate 600 dpi $= \frac{2160000 \times 8 \times 120}{60} = 34560000$ bits/s $= 34.56$ mbps

Result 300 dpi suitable because satisfy minimum pixel ...
bandwidth

c) Apply 2bit quantization

input pixels = 12, 75, 130, 200, 250

2 bit quantization = 4 levels

output levels = 0, 85, 170, 255

original	quantized	squared error
12	0	144
75	85	100
130	170	1600
200	255	3025
250	255	25

$$\text{mean (MSE)} = \frac{144+100+1600+3025+25}{5} = \frac{4894}{5} = 978.8$$

d) Recommendation & justification:

* 300 dpi satisfy minimum 2 pixel line requirement
* Its data rate 8.64mbps within 10mbps network limit
* 8 bit depth provides 256 gray level and better image quality
* lower bit depths reduce image quality may cause false contouring.
* False contouring produce visible unwanted bands & smooth grayscale regions due to insufficient gray levels.

Ceramic - Tile crack detection

a) Gin: Starting point $p = (2,3)$

Ending point $q = (6,6)$

1 Pixel = 0.5mm

Difference $\Delta r = 6-2 = 4$ $\Delta C = 6-3 = 3$

Euclidean distance $D_E = \sqrt{(\Delta r)^2 + (\Delta C)^2} = \sqrt{4^2+3^2} = \sqrt{25} = 5$

in mm = $5 \times 0.5 = 2.5$mm

city block distance: $D_4 = |\Delta r| + |\Delta C| = 4+3 = 7$ pixels

in mm $7 \times 0.5 = 3.5$mm

chessboard distane $D_8 = \max(|\Delta r|, |\Delta C|) = \max(4,3) = 4$ Pixels

$4 \times 0.5 = 2$mm

b) $p = (2,3)$

4 neighbour (N_4): $N_4(p) = \{(1,3),(3,3),(2,2),(2,4)\}$

Diagonal neighbour (N_D) = $N_D(p) = \{(1,2),(1,4),(3,2),(3,4)\}$

Adjacency justification:

* 4 adjacency - only horizontal and vertical neighbours are connected.
* 8 adjacency - includes horizontal vertical and diagonal connects diagonal crack.
* m adjacency - Avoids ambiguous connection b/w diagonal neighbours.

c) Apply power law

Gin: 20, 40, 180, 220

power law $S = 255 \times \left(\frac{r}{255}\right)^{0.5}$

input (r)	output (s)
20	71.42
40	101.00
180	214.20
220	236.85

Linear contrast stretching:

input = [20, 220] output = [0, 255]

$$S = \frac{r-20}{220-20} \times 255$$

input (r)	output (s)
20	0
40	25.5
180	204
220	255

d) calculate contrast

contrast = lowest glaze value - Highest crack value

Transformation 180 - 40 = 140

After powerlaw transformation

214.20 - 101.00 = 113.20

After linear contrast stretching:

= 204 - 25.5

= 178.5

Result: linear contrast stretching better because increase contrast 140 to 178.5 making crack easier distinguish good glaze.

Undersea pipeline inspection

a) Image size $= 8 \times 8 = 64$

Gray level = 0 to 7

Total pixels = 64

$$P_r(r_k) = \frac{n_k}{64}$$

CDF

Intensity (r_k)	Pixel count (n_k)	normalized histogram	CDF
0	4	0.0625	0.0625
1	10	0.15625	0.21875
2	16	0.25	0.46875
3	14	0.21875	0.6875
4	8	0.125	0.8125
5	6	0.09375	0.90625
6	4	0.0625	0.96875
7	2	0.03125	1.0000
Total	64	1.0000	

b) Apply histogram

$$S_k = \text{round}\,[(L-1) \times CDF]$$

$S_0 = \text{round}(7 \times 0.0625) = 0$

$S_1 = \text{round}(7 \times 0.21875) = 2$

Input level	CDF	Equilized level
0	0.0625	0
1	0.21875	2
2	0.46875	3
3	0.6875	5
4	0.8125	6
5	0.90625	6
6	0.96875	7
7		7

Equlized histogram	output level	pixel count
	0	4
	1	0
	2	10
	3	16
	4	0
	5	14
	6	14
	7	6
		64

c) calculate mean $\mu = \frac{\Sigma r_k n_k}{N}$

$= \frac{(0\times4)+(1\times10)+(2\times16)+(3\times14)+(4\times8)+(5\times6)+(6\times4)+(7\times2)}{64} = \frac{184}{64}$

$\mu = 2.875$

$\sigma = \sqrt{\frac{\Sigma n_k (r_k-\mu)^2}{N}}$ $\sigma = 1.728$ $\mu = \frac{\Sigma s_k n_k}{64} = \frac{264}{64}$ $\mu = 4.125$

$\sigma = \sqrt{\frac{\Sigma n_k (s_k-\mu)^2}{64}}$ $\sigma = 1.949$

d) compare enhancement methods

* The Rov lamp produces uneven illumination underwater
* Global histogram equalization may not enhance all regions equally.
* Linear stretching applies the same contrast adjustment to the entire image.
* AHE improves local contrast and makes small corrosion pits easier to identify.

PCB inspection

a) $$I = \begin{bmatrix} 12 & 15 & 85 \\ 10 & 14 & 90 \\ 8 & 12 & 105 \end{bmatrix}$$

$T = 200$

$G_x = (-1)(12) + (1)(85) + (-2)(10) + (2)(90) + (-1)(8) + (1)(105)$

$= -12 + 85 - 20 + 180 - 8 + 105$

$G_x = 330$

$G_y = -12 - 30 - 85 + 8 + 24 + 105$

$G_y = 10$

Gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$

$= \sqrt{330^2 + 10^2} = \sqrt{109000}$

$G = 330.15$

Gradient direction: $\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$

$= \tan^{-1}\left(\frac{10}{330}\right)$

$\theta = 1.74^\circ$

$330.15 > 200$

b) prewitt mask

$G_x = (-12 + 85) + (-10 + 90) + (-8 + 105) = 250$

$G_y = (-12 - 15 - 85) + (8 + 12 + 105) = 14$

$G = \sqrt{250^2 + 14^2} = \sqrt{62696}$

$G = 250.39$

c) Iterative global thresholding:

pixel Intensity 22, 28, 30, 36, 64, 170, 185, 195, 210

$$T_0 = \frac{\text{Sum of intensities}}{\text{Total pixels}} = \frac{940}{9} = 104.44$$

Calculate Group

Group 1: (below threshold) 22, 28, 30, 36, 64

Group 2: (above threshold) 170, 185, 195, 210

$$m_1 = \frac{22+28+30+36+64}{5} = 36 \qquad m_2 = \frac{170+185+195+210}{4} = 190$$

$$T_1 = \frac{m_1+m_2}{2} = \frac{36+190}{2} \quad T_1 = 113$$

$$T_2 = \frac{36+190}{2} = 113$$

$$|T_2 - T_1| = |113-113| = 0$$

0 < 0.5 algorithm

d) Flux residue thresholding methods:

Flux residue Intensity = 64

Final threshold = 113

64 below 113 classified board

Global thresholding: use entire image.

Otsu's method: Automatically select optimal threshold two Intensity groups.

Region growing: Group neighbouring pixels similar intensity helps identify connected regions.